Distributions and Maximum Likelihood

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

Use the method of Lagrange multipliers to find the maximum volume of a rectangular box with faces parallel to the coordinate planes inscribed in the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$.

Let the corner of the box in the first octant be (x,y,z) with $x,y,z>0$. The total volume is $V(x,y,z)=(2x)(2y)(2z)=8xyz$. The constraint is $g(x,y,z)=\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}-1=0$. 1. Set up the Lagrange multiplier system $ablaV=\lambda ablag$: $8yz=\lambda\frac{2x}{a^2}, \quad 8xz=\lambda\frac{2y}{b^2}, \quad 8xy=\lambda\frac{2z}{c^2}$. 2. Multiply the three equations by x,y,z respectively.

2 **[Foundational]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

A temperature distribution is given by $T(x,y,z)=\frac{120}{1+x^2+2y^2+3z^2}$. At the point $P(1,1,1)$, find: (a) the direction of steepest temperature increase, (b) the maximum rate of increase, and (c) the equation of the tangent plane to the isothermal level surface passing through P.

1. Compute the gradient $ablaT$: Let $u=1+x^2+2y^2+3z^2$. At $P(1,1,1)$, $u=1+1+2+3=7$. $ablaT=-\frac{120}{u^2}\langle 2x,4y,6z\rangle$. At $P(1,1,1)$: $ablaT(1,1,1)=-\frac{120}{49}\langle 2,4,6\rangle=-\frac{240}{49}\langle 1,2,3\rangle$. (a) Direction of steepest increase: $\mathbf{u}_{\max}=\frac{ablaT}{||ablaT||}=-\frac{\langle 1,2,3\rangle}{\sqrt{1+4+9}}=-\frac{1}{\sqrt{14}}\langle 1,2,3\rangle$. (b) Maximum rate of increase.

3 **[Foundational]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

Level 3 · AdvancedWater flows into an initially empty tank at a rate $R(t)=120-4t$ liters/hour and leaks out at a rate $L(t)=3t$ liters/hour for $0\le t\le 24$ hours. Find the maximum volume of water in the tank and the time at which the tank becomes completely empty.

Step 1 · Volume Accumulation Function: Net rate of change is $V'(t)=R(t)-L(t)=(120-4t)-3t=120-7t$. Since the tank is initially empty, $V(0)=0$: $V(t)=\int_0^t(120-7u)\,du=120t-\frac{7}{2}t^2$. Step 2 · Maximum Volume: Maximum occurs when $V'(t)=0\Rightarrow 120-7t=0\Rightarrow t=\frac{120}{7}\approx 17.14$ hours. Since x for $t>0$: $120t-\frac{7}{2}t^2=0\Rightarrow t(120-\frac{7}{2}t)=0\Rightarrow t=\frac{240}{7}\approx 34.29$ hours. Since t.

4 **[Foundational]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

[Surface Integrals, the Divergence Theorem & Stokes' Theorem · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

Substitute $t=8$: $g(8)=5(8)+5=\mathbf{45}$. Setting $5t+5=50$ gives $5t=45\Rightarrow t=\mathbf{9}$.

© ClipSAT — clipsat.org | All Rights Reserved

5 **[Foundational]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

[Triple Integrals & Cylindrical/Spherical Coordinates · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

6 **[Foundational]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

[Vector Fields & Line Integrals · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

7 **[Foundational]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

Let $z = f(x, y)$ be a differentiable function, with polar coordinates transformation $x = r\cos\theta$ and $y = r\sin\theta$. Express $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$ in terms of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$, and prove that: $\left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2 = \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = ||ablaz||^2$.

1. Apply the Multivariable Chain Rule: $\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r} = z_x\cos\theta + z_y\sin\theta$.

$\frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial \theta} = z_x(-r\sin\theta) + z_y(r\cos\theta) = r(-z_x\sin\theta + z_y\cos\theta)$. 2. Divide by r and square both expression.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

Find the total mass and the z-coordinate of the center of mass of the upper solid hemisphere $x^2 + y^2 + z^2 \le R^2$, $z \ge 0$, whose density is proportional to the distance from the origin: $\delta(\rho, \phi, \theta) = k\rho$ for constant $k > 0$.

1. Coordinate Setup: In spherical coordinates: $0 \le \rho \le R$, $0 \le \phi \le \frac{\pi}{2}$, $0 \le \theta \le 2\pi$, $dV = \rho^2\sin\phi\, d\rho\, d\phi\, d\theta$.

2. Total Mass M: $M = \iiint_E (k\rho)(\rho^2\sin\phi)\, d\rho\, d\phi\, d\theta = k\left(\int_0^{2\pi} d\theta\right)\left(\int_0^{\pi/2} \sin\phi\, d\phi\right)\left(\int_0^R \rho^3\, d\rho\right)$. $M = k(2\pi)(1)\left(\frac{R^4}{4}\right) = \frac{\pi k R^4}{2}$. 3. Moment with respect to xy-plane M_{xy}: Since.

9 **[Intermediate]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

Derive the Gaussian integral $I = \int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}$ from first principles by evaluating the double integral $I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2} dA$ in polar coordinates.

1. Express I^2 as a double integral: By Fubini's theorem on independent variables:

$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2}\, dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2/2}\, dy\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)/2}\, dx\, dy$. 2. Convert to polar coordinates: Let $x = r\cos\theta, y = r\sin\theta$, so $x^2 + y^2 = r^2$ and $dA = r\, dr\, d\theta$. The domain $\mathbb{R}^2$ maps to $0 \le r < \infty$ and $0 \le \theta \le 2\pi$.

© ClipSAT — clipsat.org | All Rights Reserved

10 **[Intermediate]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

Prove that if $\mathbf{x}_p$ is any particular solution to $A\mathbf{x} = \mathbf{b}$, then the complete solution set is the affine subspace $S = \{\mathbf{x}_p + \mathbf{x}_h : \mathbf{x}_h \in \text{Null}(A)\}$.

Two-part set inclusion proof: 1. Show $S \subseteq \text{Solution Set}$: Let $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$ where $A\mathbf{x}_p = \mathbf{b}$ and $A\mathbf{x}_h = \mathbf{0}$. Applying matrix multiplication: $A\mathbf{x} = A(\mathbf{x}_p + \mathbf{x}_h) = A\mathbf{x}_p + A\mathbf{x}_h = \mathbf{b} + \mathbf{0} = \mathbf{b}$. Thus every vector in S is indeed a valid solution. 2. Show $\text{Solution Set} \subseteq S$: Let $\mathbf{y}$ be an arbitrary solution to $A\mathbf{x} = \mathbf{b}$, so.

11 **[Intermediate]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

Compute the moment of inertia I_z about the z-axis of a solid homogeneous right circular cone of base radius R, height h, and uniform total mass M.

1. Cone Geometry & Density: The cone has vertex at the origin and base at $z = h$, so the lateral surface is $r = \frac{R}{h}z \Rightarrow z = \frac{h}{R}r$. Total volume $V = \frac{1}{3}\pi R^2 h$, so density $\delta_0 = \frac{3M}{\pi R^2 h}$. 2. Cylindrical Integral Setup for $I_z = \iiint_E r^2 \delta_0 \, dV$: Bounds: $0 \le \theta \le 2\pi$, $0 \le r \le R$, $\frac{h}{R}r \le z \le h$.

12 **[Intermediate]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

Find the volume and the z-coordinate of the centroid $\bar{z}$ of the solid region bounded below by the xy-plane $z = 0$ and above by the paraboloid $z = 4 - x^2 - y^2$, assuming uniform density $\delta_0 = 1$.

1. Cylindrical Coordinate Setup: The paraboloid intersects $z = 0$ at $r^2 = 4 \Rightarrow r = 2$. Bounds: $0 \le \theta \le 2\pi$, $0 \le r \le 2$, $0 \le z \le 4 - r^2$, with $dV = r\,dz\,dr\,d\theta$. 2. Volume: $V = \int_0^{2\pi}\int_0^2\int_0^{4-r^2} r\,dz\,dr\,d\theta = 2\pi\int_0^2 r(4 - r^2)\,dr = 2\pi[2r^2 - \frac{r^4}{4}]_0^2 = 2\pi(8 - 4) = 8\pi.$ 3. Moment M_{xy}.

13 **[Intermediate]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

Use the Gauss–Jordan elimination algorithm on the augmented block matrix $[A \mid I_3]$ to compute A^{-1} for: $A = (1, 0, 2; 2, -1, 3; 4, 1, 8)$.

Step 1: Set up $[A \mid I]$: $(1, 0, 2, 1, 0, 0; 2, -1, 3, 0, 1, 0; 4, 1, 8, 0, 0, 1) \to (1, 0, 2, 1, 0, 0; 0, -1, -1, -2, 1, 0; 0, 1, 0, -4, 0, 1).$ Step 2: Row swap and scaling: $\to (1, 0, 2, 1, 0, 0; 0, 1, 0, -4, 0, 1; 0, -1, -1, -2, 1, 0) \to (1, 0, 2, 1, 0, 0; 0, 1, 0, -4, 0, 1; 0, 0, -1, -6, 1, 1).$ Step 3: Clear column 3 to reach reduced row echelon form.

14 **[Intermediate]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

Evaluate the Cartesian triple integral $\iiint_B xyz^2 \, dV$ over the rectangular box $B = [0, 1] \times [-1, 2] \times [0, 3]$.

Since the region is a Cartesian box and the integrand is completely separable into univariate functions $f(x, y, z) = x \cdot y \cdot z^2$, Fubini's Theorem allows complete factoring: $\iiint_R xyz^2 \, dV = \left(\int_0^1 x\,dx\right)\left(\int_{-1}^2 y\,dy\right)\left(\int_0^3 z^2\,dz\right)$. Compute each single-variable factor: $\int_0^1 x\,dx = [\frac{x^2}{2}]_0^1 = \frac{1}{2}$. $\int_{-1}^2 y\,dy = [\frac{y^2}{2}]_{-1}^2 = \frac{4-1}{2} = \frac{3}{2}$..

© ClipSAT — clipsat.org | All Rights Reserved

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Distributions & Maximum Likelihood Practice Exercises** • **Cross-Lesson: 6.01 Linear Regression and OLS**

Express the complete solution set of the non-homogeneous linear system in parametric vector form:

$\{x_1 + 2x_2 - x_3 + 3x_4 = 4;\ 2x_1 + 4x_2 - x_3 + 8x_4 = 11;\ -x_1 - 2x_2 + 2x_3 - x_4 = -1$

Step 1: Set up augmented matrix and row reduce: $(1, 2, -1, 3, 4; 2, 4, -1, 8, 11; -1, -2, 2, -1, -1) \to (1, 2, -1, 3, 4; 0, 0, 1, 2, 3; 0, 0, 1, 2, 3)$ $\to (1, 2, 0, 5, 7; 0, 0, 1, 2, 3; 0, 0, 0, 0, 0)$. Step 2: Identify pivot and free variables: Pivots in columns 1 and 3 (x_1, x_3). Free variables are $x_2 = s$, $x_4 = t$. From RREF: $x_1 = 7 - 2s - 5t$, $x_3 = 3 - 2t$. Step 3: Parametric vector decomposition.

16 **[Challenging]** • **Target: Distributions & Maximum Likelihood Practice Exercises** • **Cross-Lesson: 7.01 Logistic Regression and Classification**

Evaluate the double integral $\iint_R x\cos(x + y)\, dA$ over the rectangular region $R = [0, \pi] \times [0, \pi/2]$.

1. Set up the iterated integral: $\int_0^{\pi}\int_0^{\pi/2} x\cos(x + y)\, dy\, dx$. 2. Integrate with respect to y: $\int_0^{\pi/2} \cos(x + y)\, dy = [\sin(x + y)]_0^{\pi/2} = \sin(x + \pi/2) - \sin x = \cos x - \sin x$. 3. Integrate with respect to x using integration by parts: $\int_0^{\pi} x(\cos x - \sin x)\, dx = \int_0^{\pi} x\cos x\, dx - \int_0^{\pi} x\sin x\, dx$. $\int_0^{\pi} x\cos x\, dx = [x\sin x]_0^{\pi} - \int_0^{\pi} \sin x\, dx = 0 - [-\cos x]_0^{\pi} = -(1 - (-1)) = -2$..

17 **[Challenging]** • **Target: Distributions & Maximum Likelihood Practice Exercises** • **Cross-Lesson: 8.01 PCA and Dimensionality Reduction**

Evaluate the non-elementary integral $\int_0^1\int_y^1 e^{x^2} dx\, dy$ by reversing the order of integration.

1. Sketch the region of integration D: The given bounds are $y \le x \le 1$ for $0 \le y \le 1$. This describes the triangle bounded by the lines $y = 0$, $x = 1$, and $y = x$. 2. Reverse the order to $dy\, dx$: In terms of vertical cross-sections, x ranges from 0 to 1, and for each fixed x, y ranges from 0 up to the line $y = x$: $\int_0^1\int_0^x e^{x^2} dy\, dx$. 3. Evaluate the inner integral: $\int_0^x e^{x^2} dy = e^{x^2}[y]_0^x = xe^{x^2}$. 4. Evaluate the outer integral: Substitute.

18 **[Challenging]** • **Target: Distributions & Maximum Likelihood Practice Exercises** • **Cross-Lesson: 1.01 Vectors and Embeddings in AI**

Find the directional derivative of $f(x, y) = x^2y - 3y^2$ at the point $P(1, 2)$ in the direction of the vector $\mathbf{v} = \langle 3, 4 \rangle$.

1. Compute the gradient vector *ablaf*: $ablaf(x, y) = \langle 2xy,\ x^2 - 6y \rangle$. At $P(1, 2)$: $ablaf(1, 2) = \langle 2(1)(2),\ 1^2 - 6(2) \rangle = \langle 4,\ -11 \rangle$. 2. Normalize the direction vector $\mathbf{v}$: $||\mathbf{v}|| = \sqrt{3^2 + 4^2} = 5 \Rightarrow \mathbf{u} = \frac{\mathbf{v}}{||\mathbf{v}||} = \left\langle \frac{3}{5},\ \frac{4}{5} \right\rangle$. 3. Compute the directional derivative.

© ClipSAT — clipsat.org | All Rights Reserved

19 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Verify Clairaut's Theorem on equality of mixed partial derivatives for the function $f(x, y) = \cos(2x - 3y) + x^2y^3$ by computing f_{xy} and f_{yx} directly.

1. Compute f_x and f_{xy}: $f_x = \frac{\partial}{\partial x}[\cos(2x - 3y) + x^2y^3] = -2\sin(2x - 3y) + 2xy^3$. $f_{xy} = \frac{\partial}{\partial y}[-2\sin(2x - 3y) + 2xy^3] = -2\cos(2x - 3y)(-3) + 6xy^2 = 6\cos(2x - 3y) + 6xy^2$. 2. Compute f_y and f_{yx}: $f_y = \frac{\partial}{\partial y}[\cos(2x - 3y) + x^2y^3] = 3\sin(2x - 3y) + 3x^2y^2$. $f_{yx} = \frac{\partial}{\partial x}[3\sin(2x - 3y) + 3x^2y^2] = 3\cos(2x - 3y)(2) + 6xy^2 = 6\cos(2x - 3y) + 6xy^2$. Since f is C^2 on $\mathbb{R}^2$.

20 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Find the shortest Euclidean distance from the point $P(3, -1, 4)$ to the hyperplane $\mathcal{H} : 2x - y + 2z = 5$ in $\mathbb{R}^3$.

Hyperplane Distance Formula: For $\mathcal{H} : \mathbf{n} \cdot \mathbf{x} = d$, the perpendicular distance from $\mathbf{x}_0$ is: $D = \frac{|\mathbf{n} \cdot \mathbf{x}_0 - d|}{||\mathbf{n}||}$. Here $\mathbf{n} = (2, -1, 2)^T$, $d = 5$, and $\mathbf{x}_0 = (3, -1, 4)^T$. 1. Norm of normal vector: $||\mathbf{n}|| = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$. 2. Value at point P: $\mathbf{n} \cdot \mathbf{x}_0 = 2(3) - 1(-1) + 2(4) = 6 + 1 + 8 = 15$. 3. Distance.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Consider the parametric curve defined by $x(t) = 3\cos(t) - \cos(3t)$ and $y(t) = 3\sin(t) - \sin(3t)$ for $t \in [0, 2\pi]$. This curve traces out a shape known as a 'cardioid'. Find the maximum area of a triangle that can be inscribed within the loop of this cardioid, with one vertex at the origin.

a. $\frac{27\sqrt{3}}{2}$

b. $\frac{81}{2}$

c. $\frac{81\sqrt{3}}{4}$ ✓

d. $54\sqrt{3}$

The parametric equations describe a cardioid. We need to find the loop of the cardioid. Let's analyze the behavior of $x(t)$ and $y(t)$. The curve forms a loop when the path self-intersects. This typically happens when the parameter values lead to the same point. To find the area of a triangle with one vertex at the origin and the other two vertices on the curve, we can use the formula for the area of a triangle given two sides and the included angle, or by using the determinant formula: Area =.

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A marketing research firm is analyzing the relationship between the preferred social media platform and age group of consumers. They collected data from a sample and summarized it in the two-way table below. The marginal distributions are also provided. The research team is interested in understanding how the distribution of preferred platform varies across age groups and is considering a chi-square test for independence. If the null hypothesis of independence is rejected, the observed counts will be significantly different from what would be expected under independence. Assume that a chi-square test for independence is to be performed on this data. Which of the following represents a potential challenge or insight derived from the structure of the data and the test's assumptions?

a. The number of degrees of freedom for the chi-square test can be determined solely from the total number of observations, irrespective of the table dimensions.

b. If the expected count in any cell is less than 5, a normal approximation to the binomial distribution for each cell count would be appropriate to proceed with the chi-square test.

c. The calculation of expected counts under the null hypothesis of independence relies on the marginal proportions of the observed data. ✓

d. A high chi-square statistic indicates that the observed counts are very close to the expected counts, suggesting a strong association between age group and preferred social media platform.

The chi-square test for independence assumes that the expected counts in each cell are sufficiently large, typically at least 5. Option A is incorrect because degrees of freedom depend on the number of rows and columns: df = (r-1)(c-1). Option B is incorrect; if expected counts are too small, alternative methods like Fisher's exact test or combining categories might be necessary, not a normal approximation for cell counts in this context. Option D is incorrect; a high chi-square statistic indicates that the observe.

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A researcher is analyzing the relationship between the number of hours a student spends on extracurricular activities per week and their final exam score in a challenging STEM course. The researcher collects data from 100 students and fits a linear regression model. The model is given by $\text{Exam Score} = 65 + 2.5 \times \text{Hours} - 0.3 \times \text{Hours}^2$, where Exam Score is the final exam score and Hours is the number of hours spent on extracurricular activities per week. The researcher believes that for a certain range of hours, there might be an optimal number of extracurricular hours that maximizes the exam score, and beyond that, the score might decrease due to time constraints. \\ \\ Consider the function representing the exam score as a function of hours: $f(h) = 65 + 2.5h - 0.3h^2$. To find the maximum exam score and the corresponding hours, calculus can be used. \\ Which of the following is the approximate number of extracurricular hours per week that would maximize the student's final exam score, assuming the model holds for $h \geq 0$?

a. **4.17 hours ✓**

b. 2.5 hours

c. 5.0 hours

d. 8.33 hours

The exam score is modeled by the quadratic function $f(h) = 65 + 2.5h - 0.3h^2$. This is a downward-opening parabola, so its maximum occurs at the vertex. To find the vertex, we can use calculus and find the critical points by taking the derivative with respect to h and setting it to zero: \\ $f'(h) = \frac{d}{dh}(65 + 2.5h - 0.3h^2) = 2.5 - 0.6h$. \\ Setting the derivative to zero to find the critical point: \\ x. \\ To confirm this is a maximum, we can check the second derivative: $f''(h) = -0.6$. Since the s.

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A pharmaceutical company is testing a new drug designed to reduce blood pressure. They decide to conduct a randomized controlled experiment. To investigate the potential for confounding variables related to lifestyle, they collect preliminary data on participants' self-reported levels of physical activity and dietary habits. They categorize physical activity into three levels: 'Low', 'Moderate', and 'High'. Dietary habits are categorized into four types: 'Vegetarian', 'Pescatarian', 'Low-Carb', and 'Omnivore'. If the company wants to ensure that any observed association between these two categorical variables in their sample data can be rigorously tested for statistical significance using a chi-square test of independence, and they obtain a contingency table with 3 rows (for physical activity) and 4 columns (for dietary habits), what is the maximum number of independent pieces of information contributing to the chi-square statistic, often referred to as the degrees of freedom, in this specific test?

a. 12

b. 9

c. **6** ✓

d. 2

The degrees of freedom (df) for a chi-square test of independence are calculated using the formula: df = (number of rows - 1) * (number of columns - 1). In this scenario, the number of rows representing physical activity levels is 3, and the number of columns representing dietary habits is 4. Therefore, df = (3 - 1) * (4 - 1) = 2 * 3 = 6. This value represents the number of cells in the contingency table whose values can be freely chosen or estimated before the remaining cell values are determined by the marginal t.

25 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

What is the maximum rate of change of $f(x, y) = x\cos(y) - 3y$ at $(2,\ 3)$?

a. $3\sin^2(3) + 12\sin(3) + 10$

b. $2\sin(3) - \cos(3) + 3$

c. $\sqrt{3\sin^2(3) + 12\sin(3) + 10}$ ✓

d. $2\sin(3) + 3$

The maximum rate is $||ablaf||$. Here $ablaf(2,\ 3) = \langle\cos(3),\ -3 - 2\sin(3)\rangle$, so it is $\sqrt{3\sin^2(3) + 12\sin(3) + 10}$.

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Maximize $f(x, y) = xy$ subject to $x + y = -4$. Find the maximum value.

a. 16

b. **4** ✓

c. 5

d. -4

$g(x, y) = x + y$, $ablag = (1, 1)$. $ablaf = (1y, 1x)$; setting $ablaf = \lambda ablag$ gives $1y = \lambda = 1x \Rightarrow x = y$. With $x + y = -4$: $x = y = -2$, so $f = 4$.

27 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Let $F(x) = \int_0^x (t^2 + 5t + 4)\, dt$. At which x does F have a local maximum?

a. -1

b. $-\frac{5}{2}$

c. 0

d. -4 ✓

$F'(x) = (x + 1)(x + 4)$. F' changes sign at -4 (from + to –) and at -1 (from – to +), so the local max is at $x = -4$.

28 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Find the maximum of $f(x, y) = xy$ on the ellipse $\frac{x^2}{4} + \frac{y^2}{25} = 1$.

a. 5 ✓

b. 10

c. $\frac{5}{2}$

d. $\frac{7}{2}$

$ablaf = \lambda ablag$ gives $y = \lambda \frac{2x}{4}$, $x = \lambda \frac{2y}{25}$, so $\frac{x^2}{4} = \frac{y^2}{25} = \frac{1}{2}$. Then $xy = 5$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

Find the global maximum value of $f(x) = x^3 - 3x^2$ on $[-1, 3]$.

a. **0** ✓

b. 3

c. 2

d. -4

Critical points where $f'(x) = 3x^2 - 6x = 0$, plus endpoints $x = -1, 3$. Evaluating f at each candidate, the largest value is $f(0) = 0$.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

Find the maximum of $f(x, y) = 5x + 12y$ on the circle $x^2 + y^2 = 25$.

a. 85

b. 325

c. 65 ✓

d. 169

The maximum occurs where $ablaf = \lambda ablag$, i.e. at $(x, y) = \frac{5}{\sqrt{169}}(5, 12)$. Then $f = 65$.

31 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

For $f(x) = x^3 - 12x$, find the local maximum value of f.

a. 14

b. **16** ✓

c. -16

d. 20

$f'(x) = 3x^2 - 12 = 0 \Rightarrow x = \pm 2$. Since $f''(x) = 6x$, $f''(-2) < 0$ (local max). $f(-2) = 16$.

32 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

A population has $\mu = 141$ and $\sigma = 30$. For a random sample of size $n = 36$, find $P(\bar{x} < 143)$ using the sampling distribution of the sample mean.

a. 0.7554

b. 0.9554

c. 0.3554

d. **0.6554** ✓

$\sigma_{\bar{x}} = \sigma/\sqrt{n} = 5$. $z = \frac{143 - 141}{5} = 0.4$. $P(\bar{x} < 143) = 0.6554$.

33 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

The Central Limit Theorem states that sample means are approximately Normal when:

a. Population is Normal

b. **n is large (n≥30)** ✓

c. σ is known

d. p<0.5

CLT: for large n, $\bar{x}$ is approximately Normal regardless of population shape.

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

A population proportion is $p = 0.38$. For samples of size $n = 300$, find the standard deviation of the sampling distribution of $\hat{p}$.

a. 0.058

b. **0.028** ✓

c. 0.018

d. 0.068

$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.38 \times 0.62}{300}} = 0.028.$

35 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

Find the maximum of $x + y + z$ on the sphere $x^2 + y^2 + z^2 = 3$.

a. 3 ✓

b. 9

c. $\sqrt{3}$

d. $3\sqrt{3}$

$ablaf = \lambda ablag$ gives $x = y = z$, so $3x^2 = 3$ and the maximum is $3x = 3$.

36 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

A Normal random variable has $\mu = 189$ and $\sigma = 8$. Find $P(173 < X < 213)$.

a. 0.8759

b. **0.9759** ✓

c. 0.7759

d. 0.6759

$z_1 = \frac{173-189}{8} = -2$, $z_2 = \frac{213-189}{8} = 3$. $P(173 < X < 213) = 0.9759$.

37 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

Maximum of xy on ellipse $x^2/4 + y^2/9 = 1$ by Lagrange:

a. 3/2 ✓

b. 3

c. 9/4

d. 6

Setting up Lagrange: $y = \lambda x/2$, $x = 2\lambda y/9 \Rightarrow xy = 3/2$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Equal squares are cut from the corners of a 24 cm × 24 cm sheet to make an open box. What is the maximum volume (cm^3)?

a. 1024 ✓

b. 2048

c. 864

d. 512

The optimum cut is $x = 4$ (from $V' = 0$), giving $V = 1024$ cm^3.

39 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

For $f(x, y) = x^2 + y^2$ on $x^2 + y^2 \le 1$: global maximum is attained at:

a. **boundary $x^2 + y^2 = 1$ (equals 1)** ✓

b. origin

c. interior critical point only

d. corner

Max on closed disk: interior min at origin; max on boundary = 1.

40 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Events A and B are independent with $P(A) = 0.33$ and $P(B) = 0.88$. Find $P(A \cap B)$.

a. 0.4904

b. **0.2904** ✓

c. 0.2404

d. 0.1904

Independence: $P(A \cap B) = P(A) \cdot P(B) = 0.33 \times 0.88 = 0.2904$.

41 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

If $f(x) = xe^{-x}$, the maximum value occurs at $x =$

a. -1

b. 0

c. 1 ✓

d. e

$f' = (1 - x)e^{-x} = 0 \Rightarrow x = 1$. $f''(1) < 0$ confirms maximum.

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Binomial distribution requires:

a. Continuous outcomes

b. **Fixed n, two outcomes, constant p, independence ✓**

c. Normal data

d. Dependent trials

BINS: Binary, Independent, fixed N, same probability p.

43 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

A population has $\sigma = 30$. For random samples of size $n = 9$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error).

a. 13

b. 6

c. 12

d. **10 ✓**

$\sigma_{\bar{x}} = \sigma/\sqrt{n} = 30/\sqrt{9} = 30/3 = 10$.

44 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Events A and B satisfy $P(A) = 0.55$, $P(B) = 0.12$, and $P(A \cap B) = 0.0$. Find $P(A \cup B)$.

a. 0.57

b. **0.67 ✓**

c. 0.82

d. 0.62

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.55 + 0.12 - 0.0 = 0.67$.

45 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

A binomial random variable has $n = 6$ trials and success probability $p = 0.25$. Find $P(X = 6)$.

a. 0.0044

b. **0.0002 ✓**

c. 0.1318

d. 0.033

$P(X = 6) = \binom{6}{6}(0.25)^6(1 - 0.25)^0 = 0.0002$.

© ClipSAT — clipsat.org | All Rights Reserved